
2019-Ph-H1-Q13

Section: Electricity

Topic: Current, PD, Power, Resistance

Question:

A 24 V battery supplies a current of 2 A to a lamp. What is the power dissipated?

Answer:

Use $P = IV = 2 \times 24 = 48 \text{ W}$

Correct Option: D (48 W)

Guidance for Students:

This is a direct application of the power formula — no need to overthink.

Revision Tips:

- Memorise $P = IV$, $P = I^2 R$, and $P = V^2 / R$
 - Check that V is **across the component** and I is **through it**
 - Units: volts × amps = watts
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